AI FOR BUSINESS • CUSTOMER EXPERIENCE

CX GUARDIAN

Autonomous Customer Experience Recovery Platform

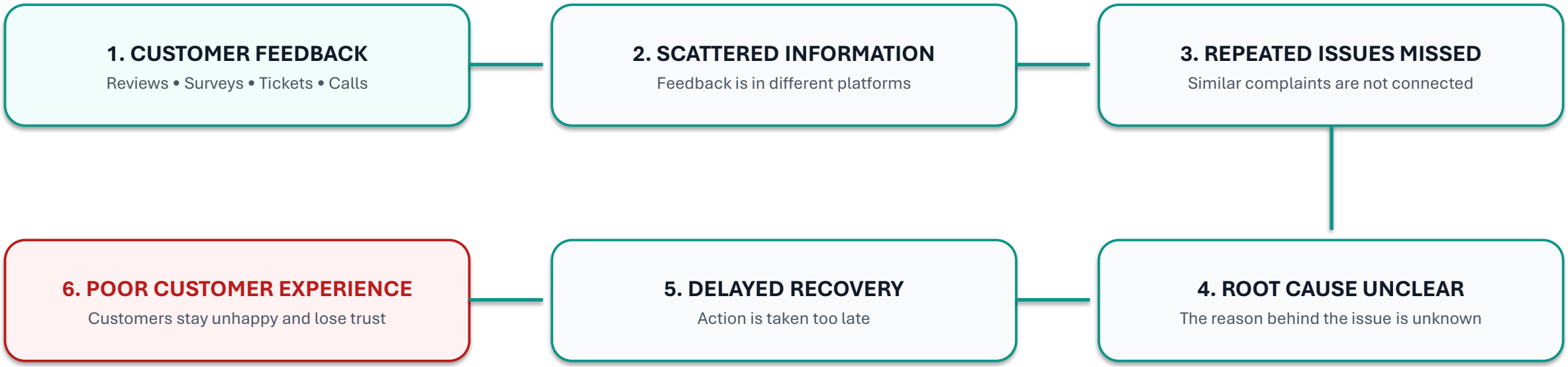
Turning scattered customer feedback into actionable insights, issue detection, and smarter recovery recommendations.

DETECT • UNDERSTAND • RECOVER

PROJECT PRESENTATION

PROBLEM STATEMENT

Businesses receive customer feedback from various sources but struggle to identify repeated complaints and their root causes. This delays problem resolution, causes recurring issues, and reduces customer satisfaction. Therefore, an intelligent system is needed to identify problems, prioritize them, and recommend timely recovery actions.

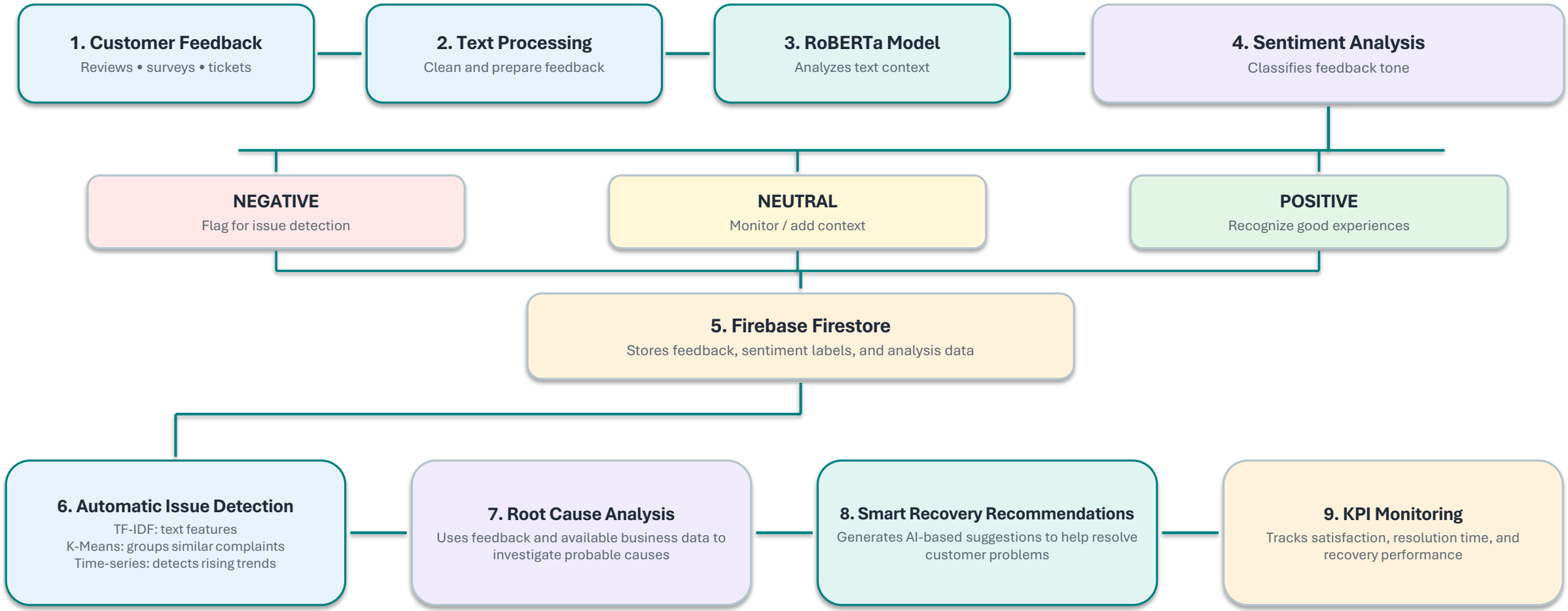


CORE PROBLEM

Customer feedback is available, but businesses struggle to connect repeated complaints to their causes and take timely action.

CX Guardian – Proposed Solution

From customer feedback to automated issue detection, recovery, and KPI monitoring



Outcome: Faster issue resolution • Fewer recurring complaints • Better customer satisfaction

Note: Sentiment indicates customer tone; root causes are investigated using additional evidence.

Technical Tools Used

CX Guardian – Autonomous Customer Experience Recovery Platform

Frontend – Website Development

React.js + Vite — Build interactive website and dashboard
Tailwind CSS — Responsive UI design and styling
Recharts — Analytics charts and graphs

Backend – Server and API

Python — AI and data processing
FastAPI — Backend REST APIs

AI / Machine Learning

Hugging Face Transformers — Load and run pretrained AI models
RoBERTa — Positive, Negative, Neutral sentiment classification
Scikit-learn — TF-IDF and K-Means clustering
Google Gemini API — Root cause assistance and recovery recommendations

Database and Authentication

Firebase Firestore — Store customer feedback, issues and recovery data
Firebase Authentication — Admin login and access control

React.js + Vite → FastAPI / Python → AI Models + Gemini API → Firebase Firestore

Technical stack grouped by frontend, backend, AI/ML, and database/authentication.

System Architecture — Application Layer

How feedback travels through the website, backend services and database

1. Feedback Sources

Reviews, surveys, support tickets and customer messages.

2. Frontend

React.js + Vite dashboard for viewing feedback, issues and KPIs.

3. Backend API

FastAPI + Python receives requests and coordinates analysis.

4. AI Processing

Calls sentiment and issue-analysis components.

5. Firestore

Stores feedback, analysis results, issue records and recommendations.

COMPONENT RESPONSIBILITIES

Frontend — React.js + Vite

Displays the dashboard, feedback records, issue summaries and charts. Sends requests to the backend API.

Backend — FastAPI + Python

Validates requests, runs processing logic, and coordinates data reads and writes.

Firestore + Authentication

Firestore stores app data. Firebase Authentication supports administrator sign-in; access rules must protect data.

Example: a team member opens a complaint in the dashboard → the frontend requests it from FastAPI → the backend retrieves the record from Firestore → the dashboard displays the feedback and its analysis.

Security note: keep privileged API keys on the backend and enforce database access permissions.

System Architecture — AI & Recovery Layer

How CX Guardian detects patterns and supports customer recovery

1. Text Preparation

Clean and prepare feedback text.

2. RoBERTa

Classifies sentiment: Positive, Neutral or Negative.

3. TF-IDF + K-Means

Represents text and groups similar complaints.

4. Trend Analysis

Checks whether complaint topics are increasing over time.

5. Gemini API

Uses feedback and available context to suggest possible causes and recovery actions.

WHAT THE ANALYSIS PRODUCES

Issue Summary

Sentiment, repeated complaint themes and rising issue trends.

Recovery Recommendation

A suggested response or service-recovery step for staff to review.

KPI Monitoring

Customer satisfaction, resolution time and recovery performance.

EXAMPLE

Several customers report late deliveries. TF-IDF + K-Means group similar complaints; trend analysis checks whether reports are increasing; Gemini suggests checking dispatch or carrier records and drafting an appropriate customer response.

Important: Gemini’s root-cause explanations are hypotheses. Staff must verify evidence before confirming a cause or taking action.

LLM Tools Used

CX Guardian — Autonomous Customer Experience Recovery Platform

PRIMARY LLM SERVICE

Google Gemini API

Generates recovery recommendations and helps investigate possible root causes using customer feedback and available business context.

SUPPORTING COMPONENTS

Prompt Engineering

Structures the instructions, context and output format given to Gemini.

RoBERTa

Classifies feedback sentiment as Positive, Neutral or Negative. It is a model, not the generative LLM.

Hugging Face Transformers

Library used to load and run the pretrained RoBERTa model.

How they work together: RoBERTa labels feedback; Gemini uses the complaint and available context to suggest next steps. Staff verify likely causes before acting.

AI / ML MODELS USED

CX Guardian | Turning customer feedback into actionable service insights

1 RoBERTa

SENTIMENT CLASSIFICATION

Classifies customer messages as Positive, Negative, or Neutral to identify dissatisfaction.

2 TF-IDF

TEXT FEATURE EXTRACTION

Converts feedback words into numerical features so text can be analyzed by machine-learning methods.

3 K-Means Clustering

COMPLAINT GROUPING

Groups similar complaints, such as late delivery, damaged products, or billing problems.

4 Time-Series Trend Analysis

TREND DETECTION

Tracks complaint counts over time and flags topics that are becoming more frequent.

Recovery support: Gemini API suggests possible causes and actions; a human reviews recommendations before action.

IMPLEMENTATION WORKFLOW

How CX Guardian turns customer feedback into actionable recovery insights

01 Collect customer feedback

Gather reviews, surveys, support tickets and customer messages.

02 Prepare and classify text

Clean the text and use RoBERTa to label sentiment as positive, neutral or negative.

03 Group related complaints

Use TF-IDF and K-Means to group similar issues, such as late delivery or billing errors.

04 Detect emerging trends

Analyze complaint counts over time to find issues that are becoming more frequent.

05 Recommend recovery actions

Gemini API suggests possible causes and responses for a human to review.

06 Monitor results

Show sentiment, complaint trends, resolution progress and recovery metrics on the dashboard.

Outcome: earlier issue detection, clearer recovery priorities, and better-informed customer service decisions.

Expected Results & Benefits

Turn customer feedback into earlier insights and more confident recovery decisions.

01

Earlier issue detection

Spot rising complaint patterns before they become larger service problems.

02

Clearer complaint insights

Group similar feedback to reveal the issues customers mention most.

03

Faster decision support

Summarize sentiment and trends so teams can prioritize what needs attention.

04

Suggested recovery actions

Use Gemini to propose responses and next steps for staff review.

05

Track service performance

Monitor complaint volume, sentiment, and recovery progress on a dashboard.

06

More consistent responses

Support a more structured approach to customer recovery across cases.

EXPECTED OUTCOME

A data-informed workflow to detect customer problems earlier, prioritize recovery, and evaluate service performance.

AI suggestions are reviewed by staff

Conclusion & Future Scope

A practical step toward faster, more informed customer recovery.

CONCLUSION

CX Guardian turns scattered customer feedback into structured, actionable insights.

Sentiment classification, complaint clustering, and trend analysis help teams identify recurring issues and prioritize attention.

Gemini can suggest recovery actions, while staff review recommendations before taking action.

Insight → review → recovery → measurement

FUTURE SCOPE

01 **Live feedback integrations**

Connect support tickets, reviews, surveys, and chat channels.

02 **Multilingual sentiment**

Support feedback in more languages and regional expressions.

03 **Smarter root-cause analysis**

Combine feedback with delivery, product, and service data.

04 **Automated alerts & tracking**

Notify teams about rising issues and follow recovery outcomes.

Goal: help teams detect issues earlier, recover trust faster, and continuously improve customer experience.